

HKCEE 1993 Mathematics II

93 1. If $f(x) = 10^{2x}$, then $f(4y) =$

- A. 10^{4y} .
- B. 10^{2+4y} .
- C. 10^{8y} .
- D. 40^y .
- E. 40^{2y} .

93 2. If $s = \frac{n}{2}[2a + (n-1)d]$, then $d =$

- A. $\frac{2(s-an)}{n(n-1)}$.
- B. $\frac{2(s-an)}{(n-1)}$.
- C. $\frac{s}{n(n-1)}$.
- D. $\frac{as-n}{a(n-1)}$.
- E. $\frac{4(s-an)}{n(n-1)}$.

93 3. Simplify $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)$.

- A. $x^4 + 1$
- B. $x^4 - x^2 + 1$
- C. $x^4 + x^2 + 1$
- D. $x^4 - 3x^2 - 2\sqrt{3}x - 1$
- E. $x^4 + 3x^3 - 2\sqrt{3}x^2 + \sqrt{3}x - 1$

93 4. $\frac{\sqrt{a}}{\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$.

- A. $\frac{1}{\sqrt{a}-\sqrt{b}}$
- B. $\frac{a+2\sqrt{ab}-b}{a-b}$

C. $\frac{\sqrt{b}+\sqrt{a}}{2\sqrt{a}}$

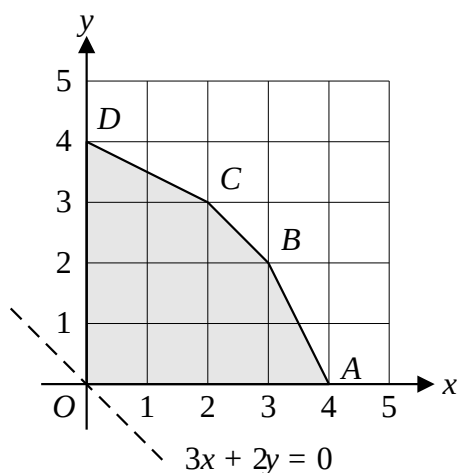
D. $\frac{b+2\sqrt{ab}-a}{a-b}$

E. $\frac{a+b}{a-b}$

93 5. If $3x^2 + ax - 5 \equiv (bx - 1)(2 - x) - 3$, then

- A. $a = -5, b = -3$.
- B. $a = -5, b = 3$.
- C. $a = -3, b = -5$.
- D. $a = 5, b = -3$.
- E. $a = 3, b = 5$.

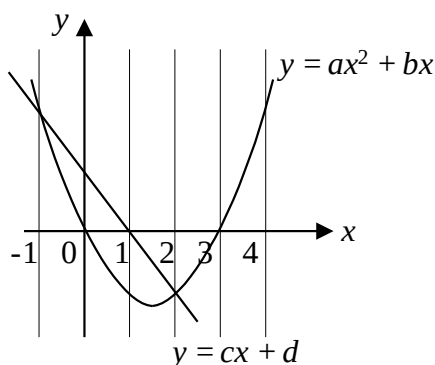
93 6.



Find the greatest value of $3x + 2y$ if (x, y) is a point lying in the region $OABCD$ (including the boundary).

- A. 15
- B. 13
- C. 12
- D. 9
- E. 8

93
7.



The diagram shows the graphs of $y = ax^2 + bx$ and $y = cx + d$. The solutions of the equation $ax^2 + bx = cx + d$ are

- A. $-1, 1$
- B. $-1, 2$
- C. $0, 1$
- D. $0, 3$
- E. $1, 3$

93
8. If $\log(p + q) = \log p + \log q$, then

- A. $p = q = 1$.
- B. $p = \frac{q}{q-1}$.
- C. $p = \frac{q}{q+1}$.
- D. $p = \frac{q+1}{q}$.
- E. $p = \frac{q-1}{q}$.

93
9. The expression $x^2 - 2x + k$ is divisible by $(x + 1)$. Find the remainder when it is divided by $(x + 3)$.

- A. 1
- B. 4
- C. 12
- D. 16
- E. 18

93
10. If 3, a , b , c , 23 are in A.P., then $a + b + c =$

- A. 13.
- B. 26.
- C. 33.
- D. 39.
- E. 65.

93
11. Find the H.C.F. and L.C.M. of ab^2c and abc^3

- | | H.C.F. | L.C.M. |
|----|-------------|-------------|
| A. | a | $a^2b^3c^4$ |
| B. | abc | ab^2c^3 |
| C. | abc | $a^2b^3c^4$ |
| D. | ab^2c^3 | abc |
| E. | $a^2b^3c^4$ | abc |

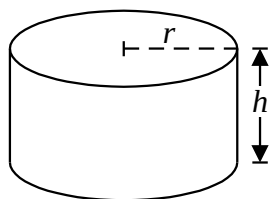
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12. If α and β are the roots of the quadratic equation $x^2 - 3x - 1 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

- A. -3
- B. -1
- C. $-\frac{1}{3}$
- D. $\frac{2}{3}$
- E. 3

93
13. If the simultaneous equations $\begin{cases} y = x^2 - k \\ y = x \end{cases}$ have only one solution, find k .

- A. -1
- B. $-\frac{1}{4}$
- C. -4
- D. $\frac{1}{4}$
- E. 1

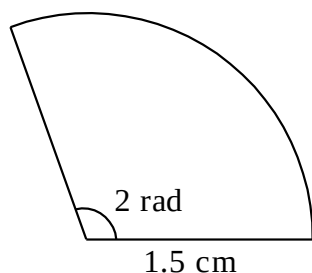
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14.



The price of a cylindrical cake of radius r and height h varies directly as the volume. If $r = 5$ cm and $h = 4$ cm, the price is \$30. Find the price when $r = 4$ cm and $h = 6$ cm.

- A. \$25
- B. \$28.80
- C. \$31.50
- D. \$36
- E. \$54

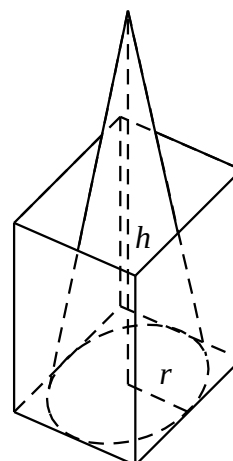
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15.



Find the perimeter of the sector in the figure.

- A. 2.25 cm
- B. 3 cm
- C. $\left(\frac{\pi}{60} + 3\right) \text{ cm}$
- D. 4.5 cm
- E. 6 cm

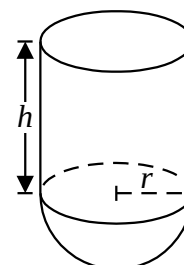
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16.



In the figure, the base of the conical vessel is inscribed in the bottom of the cubical box. If the box and the conical vessel have the same capacity, find $h : r$.

- A. $24 : \pi$
- B. $3 : 1$
- C. $6 : \pi$
- D. $3 : \pi$
- E. $8 : 3\pi$

93
17.



The figure shows a solid consisting of a cylinder of height h and a hemisphere of radius r . The area of the curved surface of the cylinder is twice that of the hemisphere. Find the ratio volume of cylinder : volume of hemisphere

- A. $1 : 3$
- B. $2 : 3$
- C. $3 : 4$
- D. $3 : 2$
- E. $3 : 1$

- 93 A merchant marks his goods 25%
18. above the cost. He allows 10 % discount on the marked price for a cash sale. Find the percentage profit the merchant makes for a cash sale.

- A. 12.5%
B. 15%
C. 22.5%
D. 35%
E. 37.5%

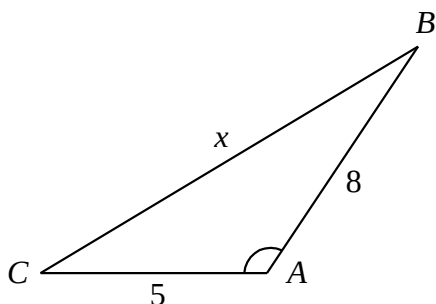
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19. $\frac{\cos \theta}{1 - \sin^2 \theta} \cdot \frac{1 - \cos^2 \theta}{\sin \theta} =$

- A. $\sin \theta$
B. $\cos \theta$
C. $\tan \theta$
D. $\frac{1}{\sin \theta}$
E. $\frac{1}{\cos \theta}$

93
20. $\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta =$

- A. 0
B. 1
C. $(1 - \sin^2 \theta)^2$
D. $(1 - \cos^2 \theta)^2$
E. $(\cos^2 \theta - \sin^2 \theta)^2$

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21.



In the figure, $\cos A = -\frac{4}{5}$. Find a .

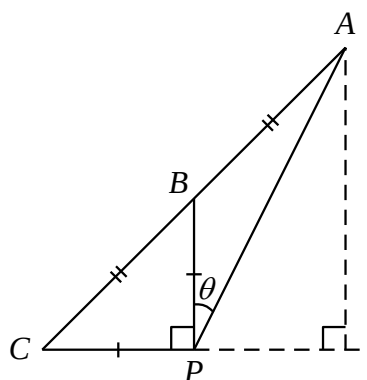
- A. $\sqrt{153}$
B. $\sqrt{137}$

- C. $\sqrt{89}$
D. $\sqrt{41}$
E. $\sqrt{25}$

93 The largest value of $3\sin^2 \theta + 2\cos^2 \theta -$
22. 1 is

- A. 1.
B. $\frac{3}{2}$.
C. 2.
D. 3.
E. 4.

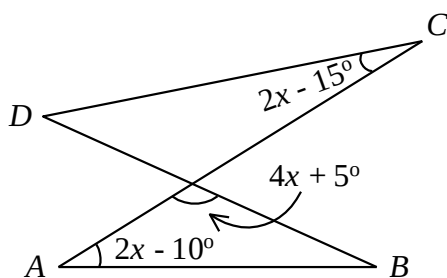
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23.



In the figure, $AB = BC$, $BP = CP$ and $BP \perp CP$. Find $\tan \theta$.

- A. $\frac{1}{4}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{\sqrt{3}}$
E. $\frac{\sqrt{3}}{2}$

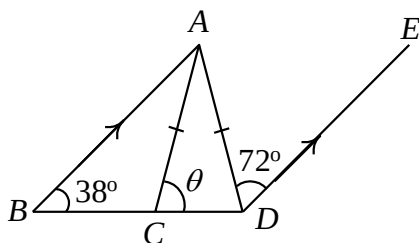
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24.



In the figure, points A , B , C and D are concyclic. Find x .

- A. 20°
- B. 22.5°
- C. 25°
- D. 27.5°
- E. 30°

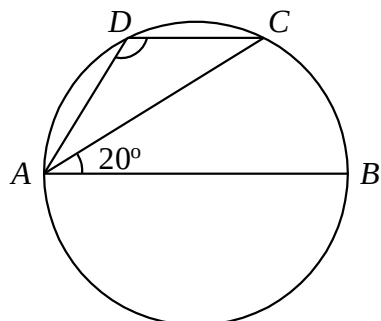
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25.



In the figure, $BA \parallel DE$ and $AC = AD$. Find θ .

- A. 34°
- B. 54°
- C. 70°
- D. 72°
- E. 76°

93
26.



In the figure, AB is a diameter. Find $\angle ADC$.

- A. 100°

- B. 110°
- C. 120°
- D. 135°
- E. 140°

93 27. If the points $(1, 1)$, $(3, 2)$ and $(7, k)$ are on the same straight line, then $k =$

- A. 3.
- B. 4.
- C. 6.
- D. 7.
- E. 10.

93 28. $A(0, 0)$, $B(5, 0)$ and $C(2, 6)$ are the vertices of a triangle. $P(9, 5)$, $Q(6, 6)$ and $R(2, -9)$ are three points. Which of the following triangles has/have area(s) greater than the area of $\triangle ABC$?

- I. $\triangle ABP$
- II. $\triangle ABQ$
- III. $\triangle ABR$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

93 29. A circle of radius 1 touches both the positive x -axis and the positive y -axis. Which of the following is/are true?

- I. Its centre is in the first quadrant.
- II. Its centre lies on the line $x - y = 0$.
- III. Its centre lies on the line $x + y = 1$.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

93 30. What is the area of the circle $x^2 + y^2 - 10x + 6y - 2 = 0$?

- A. 32π

- B. 34π
- C. 36π
- D. 134π
- E. 138π

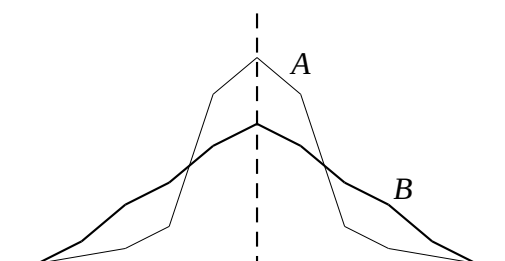
93 Two fair dice are thrown. What is the probability of getting a total of 5 or 10?

- A. $\frac{1}{9}$
- B. $\frac{5}{36}$
- C. $\frac{1}{6}$
- D. $\frac{7}{36}$
- E. $\frac{2}{9}$

93 A group of n numbers has mean m . If the numbers 1, 2 and 6 are removed from the group, the mean of the remaining $n - 3$ numbers remains unchanged. Find m .

- A. 1
- B. 2
- C. 3
- D. 6
- E. $n - 3$

93
33.



The figure shows the frequency polygons of two symmetric distributions A and B with the same mean. Which of the following is/are true?

- I. Interquartile range of A < Interquartile range of B
- II. Standard deviation of A >

Standard deviation of B
III. Mode of A > Mode of B

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

93 If $9^{x+2} = 36$, then $3^x =$
34.

- A. $\frac{2}{3}$
- B. $\frac{4}{3}$
- C. 2
- D. $\sqrt{6}$
- E. 9

93 If $a : b = 2 : 3$ and $b : c = 5 : 3$, then
35. $\frac{a+b+c}{a-b+c} =$

- A. -2
- B. $\frac{5}{2}$
- C. 4
- D. $\frac{17}{2}$
- E. 31

93
36.

x	Sign of $f(x)$
3.56	+
3.58	-
3.57	+
3.575	+

From the table, a root of the equation $f(x) = 0$ is

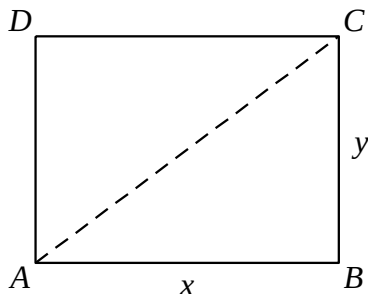
- A. 3.57 (correct to 3 sig. fig.).
- B. 3.575 (correct to 4 sig. fig.).
- C. 3.5775 (correct to 5 sig. fig.).
- D. 3.5725 (correct to 4 sig. fig.).
- E. 3.58 (correct to 3 sig. fig.).

- 93 Given that the positive numbers p, q, r, s are in G.P., which of the following must be true?

- I. kp, kq, kr, ks are in G.P., where k is a non-zero constant.
 II. a^p, a^q, a^r, a^s are in G.P., where a is a positive constant.
 III. $\log p, \log q, \log r, \log s$ are in A.P.

- A. I only
 B. II only
 C. I and II only
 D. I and III only
 E. I, II and III only

93
38.



In the figure, the rectangle has perimeter 16 cm and area 15 cm^2 . Find the length of its diagonal AC.

- A. $\sqrt{32}$ cm
 B. $\sqrt{34}$ cm
 C. 7 cm
 D. $\sqrt{226}$ cm
 E. $\sqrt{241}$ cm

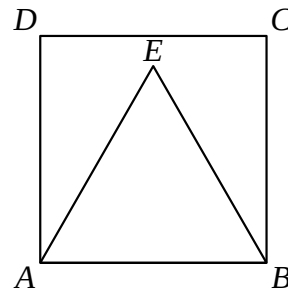
- 93 In factorizing the expression
 39. $a^4 + a^2b^2 + b^4$, we find that

- A. $(a^2 - b^2)$ is a factor.
 B. $(a^2 + b^2)$ is a factor.
 C. $(a^2 - ab - b^2)$ is a factor.
 D. $(a^2 - ab + b^2)$ is a factor.
 E. it cannot be factorized.

- 93 If the solution of the inequality
 40. $x^2 - ax + 6 \leq 0$ is $c \leq x \leq 3$, then

- A. $a = 5, c = 2$.
 B. $a = -5, c = 2$.
 C. $a = 5, c = -2$.
 D. $a = 1, c = -2$.
 E. $a = -1, c = 2$.

93
41.

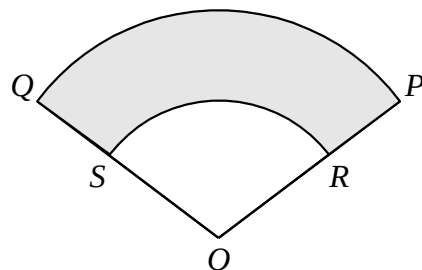


In the figure, ABCD is a square and ABE is an equilateral triangle.

$$\frac{\text{Area of } ABE}{\text{Area of } ABCD} =$$

- A. $\frac{1}{4}$
 B. $\frac{1}{3}$
 C. $\frac{\sqrt{3}}{8}$
 D. $\frac{\sqrt{3}}{4}$
 E. $\frac{\sqrt{3}}{2}$

93
42.



In the figure, the radii of the sectors OPQ and ORS are 5 cm and 3 cm respectively.

$$\frac{\text{Area of shaded region}}{\text{Area of sector } OPQ} =$$

- A. $\frac{4}{25}$.
 B. $\frac{2}{5}$.
 C. $\frac{9}{25}$.
 D. $\frac{16}{25}$.
 E. $\frac{21}{25}$.

93 43. Which of the following gives the compound interest on \$ 10 000 at 6% p.a. for one year, compounded monthly?

- A. $\$ 10\,000 \times \frac{0.06}{12} \times 12$
 B. $\$ 10\,000(1.06^{12} - 1)$
 C. $\$ 10\,000 \left(1 + \frac{0.06}{12}\right)^{12}$
 D. $\$ 10\,000 \left[\left(1 + \frac{0.06}{12}\right)^{12} - 1\right]$
 E. $\$ 10\,000 \left[\left(1 + \frac{0.6}{12}\right)^{12} - 1\right]$

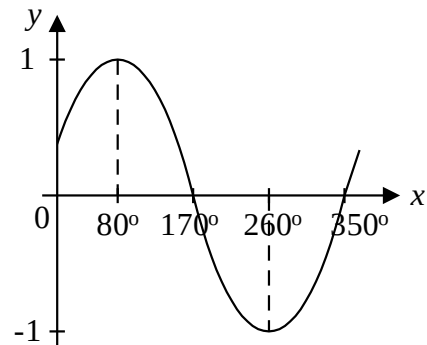
93 44. Originally $\frac{2}{3}$ of the students in a class failed in an examination. After taking a re-examination, 40% of the failed students passed. Find the total pass percentage of the class.

- A. $26\frac{2}{3}\%$
 B. $33\frac{1}{3}\%$
 C. 40%
 D. 60%
 E. $73\frac{1}{3}\%$

93 45. Solve $\tan^4 \theta + 2\tan^2 \theta - 3 = 0$ for $0^\circ \leq \theta < 360^\circ$.

- A. $45^\circ, 135^\circ$ only
 B. $45^\circ, 225^\circ$ only
 C. $45^\circ, 60^\circ, 225^\circ, 240^\circ$
 D. $45^\circ, 120^\circ, 225^\circ, 300^\circ$
 E. $45^\circ, 135^\circ, 225^\circ, 315^\circ$

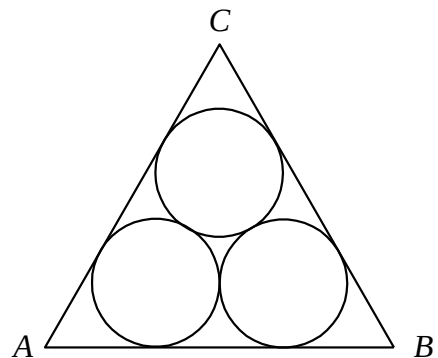
93 46.



The figure shows the graph of the function

- A. $y = \sin(350^\circ - x)$.
 B. $y = \sin(x + 10^\circ)$.
 C. $y = \cos(x + 10^\circ)$.
 D. $y = \sin(x - 10^\circ)$.
 E. $y = \cos(x - 10^\circ)$.

93 47.

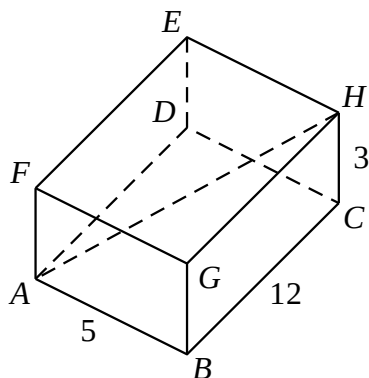


In the figure, ABC is an equilateral triangle and the radii of the three circles are each equal to 1. Find the perimeter of the triangle.

- A. 12
 B. $3(1 + \tan 30^\circ)$
 C. $6(1 + \tan 30^\circ)$

- D. $3\left(1 + \frac{1}{\tan 30^\circ}\right)$
 E. $6\left(1 + \frac{1}{\tan 30^\circ}\right)$

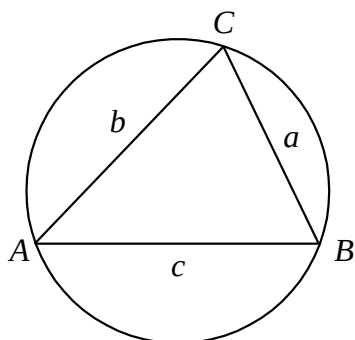
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48.



In the figure, $ABCDEFGH$ is a cuboid. The diagonal AH makes an angle θ with the base $ABCD$. Find $\tan \theta$.

- A. $\frac{3}{5}$
 B. $\frac{3}{12}$
 C. $\frac{3}{13}$
 D. $\frac{3}{\sqrt{178}}$
 E. $\frac{\sqrt{153}}{5}$

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49.



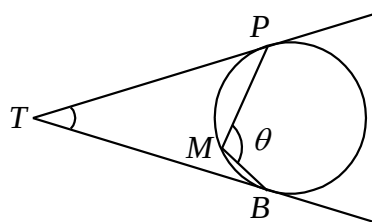
In the figure, if $\text{arc } BC : \text{arc } CA : \text{arc } AB = 1 : 2 : 3$, which of the following is/are true?

- I. $\angle A : \angle B : \angle C = 1 : 2 : 3$

- II. $a : b : c = 1 : 2 : 3$
 III. $\sin A : \sin B : \sin C = 1 : 2 : 3$

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. I, II and III only

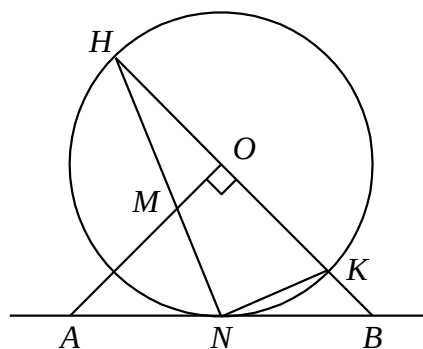
93
50.



In the figure, TP and TQ are tangent to the circle at P and Q respectively. If M is a point on the minor arc PQ and $\angle PMQ = \theta$, then $\angle PTQ =$

- A. $\frac{\theta}{2}$
 B. $\theta - 90^\circ$
 C. $180^\circ - \theta$
 D. $180^\circ - 2\theta$
 E. $2\theta - 180^\circ$

93
51.



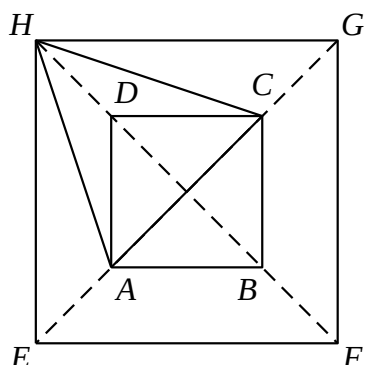
In the figure, O is the centre of the circle. AB touches the circle at N . Which of the following is/are correct?

- I. M, N, K, O are concyclic.
 II. $\triangle HNB \sim \triangle NKB$
 III. $\angle OAN = \angle NOB$

- A. I only
 B. II only

- C. III only
D. I and II only
E. I, II and III only

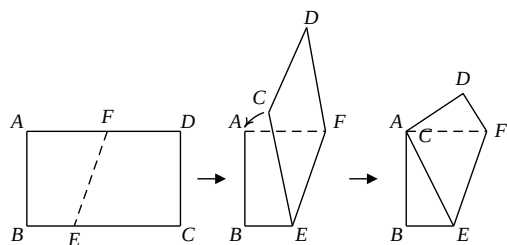
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52.



In the figure $ABCD$ and $EFGH$ are two squares and ACH is an equilateral triangle. Find $AB : EF$.

- A. $1 : 2$
B. $1 : 3$
C. $1 : \sqrt{2}$
D. $1 : \sqrt{3}$
E. $\sqrt{2} : \sqrt{3}$

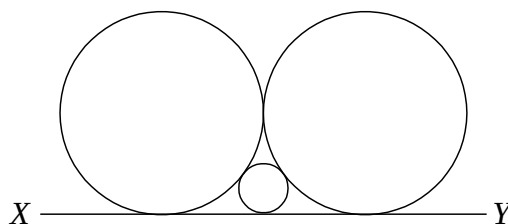
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53.



In the figure, a rectangular piece of paper $ABCD$ is folded along EF so that C and A coincide. If $AB = 12$ cm, $BC = 16$ cm, find BE .

- A. 3.5 cm
B. 4.5 cm
C. 5 cm
D. 8 cm
E. 12.5 cm

93
54.



In the figure, the three circles touch one another. XY is their common tangent. The two larger circles are equal. If the radius of the smaller circle is 4 cm, find the radii of the larger circles.

- A. 8 cm
B. 10 cm
C. 12 cm
D. 14 cm
E. 16 cm